Module	**ITC 6110 - Natural Language Processing (NLP)**
Term	**Spring Semester 2026**
Assessment	**Group Project** **Weight: 50%**
Duration	*You can use the duration of the Term to complete the project*
Deliverables	1. Report (document) submitted via Turnitin 2. Code on Blackboard or GitHub 3. An oral presentation of your work
Method of Submission	Turnitin, Blackboard, GitHub
Deadline	*13th Week* *US Grading scale*

General Instructions

Your project involves generating a series of experiments, extracting findings and observations from your experiments, and drawing conclusions. Essentially you will need to collect data and use Python as the programming language along with any appropriate libraries to process the data, generate graphs, implement models, and test results. Tables, diagrams, and data visualizations are essential for presenting your findings.

Deliverables:

- Python code submitted on Blackboard or uploaded on GitHub, along with any optional instructions for running it
- a report of 5,000±500 words that will present your findings, which will be submitted via TurnitIn. The report must be self-contained: all experiments performed along with all conclusions drawn should be reported. If you need to exceed the word limit, use an appendix.
- an oral presentation, where all members of the team need to present.

> There is zero tolerance on plagiarism, and LLMs like ChatGPT, Gemini, etc. are NOT allowed for the solutions and report.

Team size: 2-4 people (sent to the instructor by email before Week 6)

Grading

> The Grading will be uniform for all team members, based on the following breakdown.

Project guidelines

The objective of this project is to illustrate the practical usage of a range of techniques throughout the NLP pipeline. This project provides hands-on experience in NLP, covering all the various stages from data preprocessing to model development and testing. It also offers opportunities for further exploration and enhancement across various techniques (from simplistic to advanced). Your solutions should adhere to the following steps and guidelines in order to be marked. Grading will also be relevant to the effort, level of difficulty and advancements made in each step of the process. All steps outlined are mandatory unless explicitly stated as optional or extra.

Contents

1 Data Collection (10%)

Gather a dataset tailored to meet the requirements of the group project. The dataset needs to be able to address the model building exercises outlined below in Step 4.

- The dataset can originate from any public or private source*, and can feature any type of textual data (for example, it can cover movie reviews, ratings, news, articles, among others). You will need to describe your data in your report and presentation.
- The data should be in English. If the raw data contain other or multiple languages, extra pre-processing & data handling steps may need to be performed.
- The dataset needs to be labelled to conduct supervised learning tasks, specifically classification (such as sentiment analysis or topic classification; sentiment analysis is considered a subset of topic classification).
- You may need to experiment with and adjust, if needed, the size of your data, while maintaining adequate historicity and diversity, to ensure it will run smoothly on your machines without any problems. Any adjustments must be explained in your report.

****If the dataset originates from a private source, you need to get the nescecary permission from the owner of the data. Also, for all datasets, you need to comply with the utlized libraries and LLMs' terms of use.**

2 Data Preprocessing and Normalization (10%)

- As with any other ML pipeline, deal with missing and duplicate values, inconsistencies, input errors, text formats, outliers, etc.

- Normalize your input text data. Apply, where appropriate, the removal of stop words, punctuation, and special characters. Use regular expressions (RegEx) or other techniques for removal of consecutive letters, short words, etc. If in existence, handle appropriately any URLs, HTML tags, emojis, emoticons, usernames, etc. If the data contain personal, PII or sensitive information, it also needs to be removed, handled or anonymized. Experiment with and decide upon using (or not) stemming or lemmatization. Finally, apply tokenization prior to converting the data into numerical format for the next step.
- Justify all your pre-processing steps in the report, code and presentation.

3 Feature Engineering (Embeddings) & Text Visualization (15%)

- Extract features from the text data: this step involves converting the text data into numeric format. This can be achieved using various techniques like Bag of Words, TF-IDF, or word embeddings (Word2Vec, GloVe), Deep Learning techniques (e.g. Transformers) and more. Justify your choice of embedding(s) in your report and presentation. (Optional) experiment also with N-grams.
- Build a solution where you provide a word from your corpus that returns the N most similar words (5-10 words will suffice). There are multiple solutions and techniques to address this.
- Visualize the text embeddings in a scatterplot using techniques such as t-SNE (preferrable) and/or PCA. Colour the findings appropriately or add supporting text information or labels in the plots to detect any groups/clusters, patterns, trends, similarities, etc.
- (Extra points/Optional) Ideally, compare multiple embedding techniques; discuss their results, overall performance, similarities, dissimilarities, advantages and disadvantages.
- (Extra points/Optional) Feel free to use any other static or interactive visualization techniques of your choice to depict and discuss your data and findings (e.g. word frequencies).

4 Model Building (40%)

4.1 Unsupervised Learning (10%)

- Topic modelling: Apply (unsupervised learning) topic modelling using at least one algorithm of your choice to identify topic clusters, groups with similar words and hidden semantic patterns within the body of your text.
- Create supporting visualizations, extract key words per cluster (in a tabular format or visualization), and discuss your findings.

- Ideally, you can perform a comparison between various Unsupervised Learning (e.g. Multidimensional Scaling, Latent Dirichlet Allocation, etc.) techniques and discuss the results.

4.2 Supervised Learning (30%)

- Task 1 (10%): Text classification (topic classification or sentiment analysis):
  - Conduct text classification (the label can be a topic such as spam / ham or drama / comedy or sentiment such as positive / negative) using at least (a) one traditional ML technique and (b) one DL model (can be a custom solution, a pre-trained model with transfer learning, etc.). Compare the models and discuss the results.
  - Apply XAI (Explainable AI) techniques (e.g., LIME, SHAP) to explain your findings for one of your classification models on either a local or global basis.
- Task 2 (10%): Retrieval-Augmented Generation (RAG) for Conversational AI
  - Adapt the dataset used in Task 1 for retrieval-based question-answering.
  - Implement a document retriever using vector embeddings (e.g., Word2Vec, Hugging Face Embeddings)
  - Implement appropriate prompt engineering for the RAG system.
  - Use a pre-trained LLM for response generation.
  - Evaluate the performance of the RAG system.
- Task 3 (10%): UI Development with Streamlit & Deployment on Hugging Face
  - Develop an interactive user interface using Streamlit, Gradio or another UI library to allow users to input queries and receive responses from the RAG-based chatbot.
  - Deploy the model as an online application using Hugging Face Spaces or another cloud-based service.
  - Add appropriate UI elements to the application to allow users to interact with the RAG system.
  - Evaluate the user experience and model response quality, and discuss potential improvements.

5 Report quality (15%)

Document the entire project, including the dataset used, the challenges faced, preprocessing techniques investigated, model architecture, training process, deployment steps and performance metrics.

The quality of report is based on many factors including: organization of the material, presentation of data, experiments, models, evaluation, drawing conclusion using various aids such as tables, diagrams, equations etc., and references to external sources and publications. Finally, discuss potential avenues for future work.

6 Presentation (10%)

Prepare a presentation to showcase the project and its outcomes. During the presentation each group will present their work in a comprehensive manner and will be called to answer questions regarding their work. Every member in the team has to present a part of the analysis and findings.

Grading scale: US System

	GP	**Letter**	**US**
Excellent	4.00	A	95-100
Very good	3.70	A-	90-94
Very good	3.50	B+	86-89
Good	3.00	B	80-85
Satisfactory	2.50	C+	76-79
Satisfactory	2.00	C	70-75
Fail	0	F	<70

Table 1: Grading System